



Fun With Robots!

Fall 2026

Your Human Leaders



Matthew Kibarian



Nick Yaeger



Mithun Roy (TA)

Your Human Leaders

Matthew Kibarian

From: Los Altos, CA

Major: ECE, MSE Minor

Semesters teaching FWR: 2

Robot things:

- FRC (team 751: mechanical design, 3 years; Vendor product development, 1 year)
- RoboClub Training Officer
- FWR Student (F24), TA (S25)

Nick Yaeger

From: Katy, TX

Major: ECE

Semesters teaching FWR: 2

Robot things:

- CMR (Carnegie Mellon Racing) Grounded Low Voltage Team
- Roboclub Senior Member
- FWR Student (F24), TA (S25)
- 15-440 TA

Mithun Roy (TA)

From: Northfield, NJ

Major: ECE

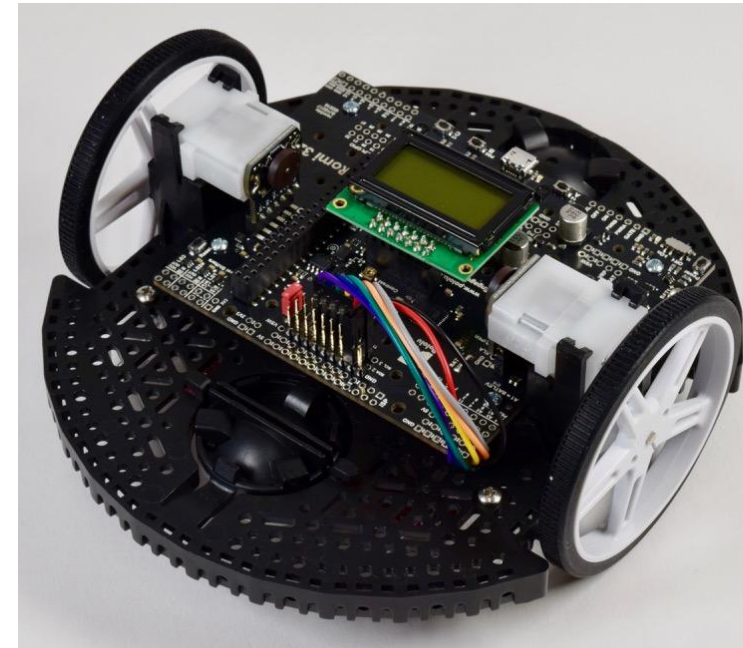
Semesters teaching FWR: 0

Robot things:

- FTC 7149
- Roboclub Asst. E-Master
- 18578 Hexapod Project
- 16-878 TA

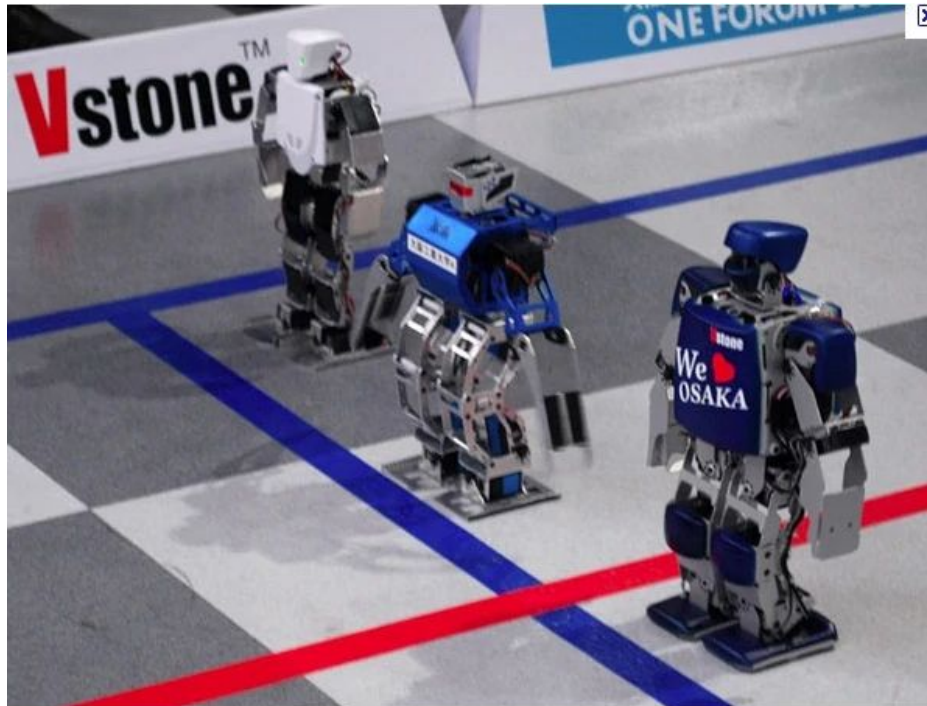
Course Plan

- Fun!
- With Robots!
- Square Lab! (Intro to robots and basic programming)
- Photovore! (Sensors)
- Line Following! (Control)
- Actuation Lab! (Servos)
- Pathfinder Lab! (Comprehensive)
- More!



Class Structure

- Short lecture (most weeks)
- Work time!
 - With candy B)
- End of a lab: compete with classmates!



https://www.google.com/search?hl=en&tbm=isch&source=hp&biw=1242&bih=658&ei=f0pkXcLkO6Wd_Qb7rLHACw&q=robot+race&og=robot+race&gs_l=img_3..0l5j0i5i30j5.1836.2557..0.0.0.73.665.10.....0....1..qws-wiz-img.AGAnW30UBiU&ved=0ahUKEwiCoJ2QuaHkAhWITt8KHxiWDLqQ4dUDCAU&uact=5#imgrc=Z-ECSI0IEVoo1M

Silly Bus

- Attendance
- Lab Fee
- Grading
- Expectations
- Other boring stuff...



Attendance

- Attendance will be taken by an attendance quiz
 - Code word will be shown every lecture
 - Submit the attendance quiz by the end of class
- StuCo Policy: > 2 unexcused absences = fail :(
 - **PLEASE** let us know if you can't come (so that we can excuse your absence)
 - Being busy / studying for exams is not a valid excuse—please plan ahead!
- Tardiness counts as an absence (30 min or more)
- You must stay for the entire scheduled period

Lab Fee

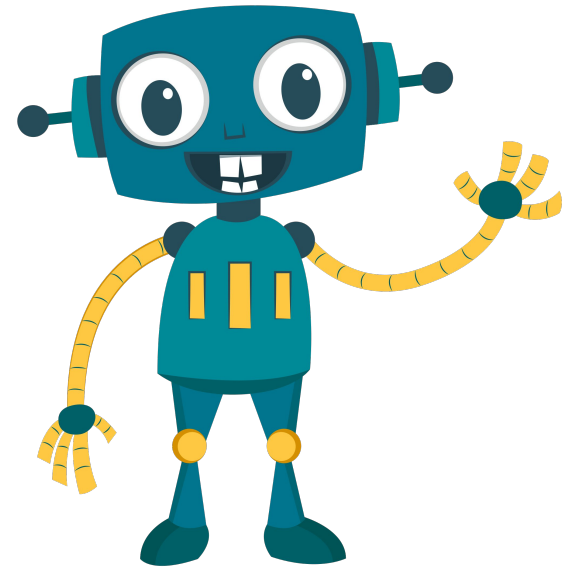
- There is a \$10 lab fee
 - To support course expenses
- You can pay with cash (preferred) or Zelle/Venmo
 - Please complete by September 14th

Survey!

<https://forms.gle/g77SgVxgEF3T3Uvt6>

This survey helps us gauge how well
we teach you throughout the
semester!

Please fill it out so we can get better!



Grading

- Labs are graded pass/fail
 - Need to pass **four (4) out of 5** labs
 - Can do three labs and an optional lab
- To pass a lab
 - Demonstrate that your robot completes the task

Links

Access lecture slides, lab slides, virtual OH, chat with peers/instructors, and more...

Discord Server



discord.gg/e2RsFzUMMG

Website



<https://www.funwithrobots.org/>



The Labs

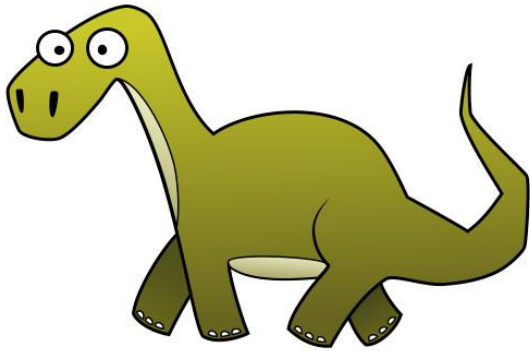
Square Lab

- Send commands to make your robot move!
 - Eventually, in a square!
- Then, robot dance competition!
 - Twerking robots, Cha Cha Slide robots, robots singing and dancing, robot waltzes....the sky's the limit!

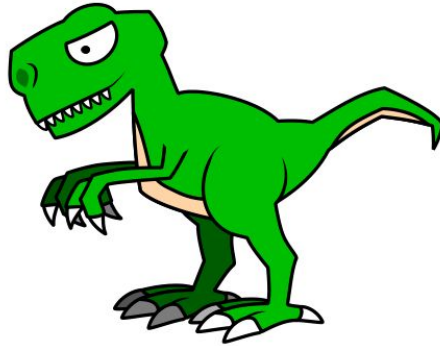


Photovore Lab

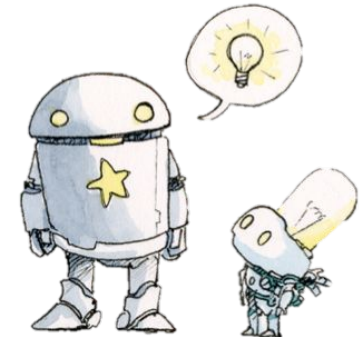
- Herbivore



- Carnivore



- Photovore



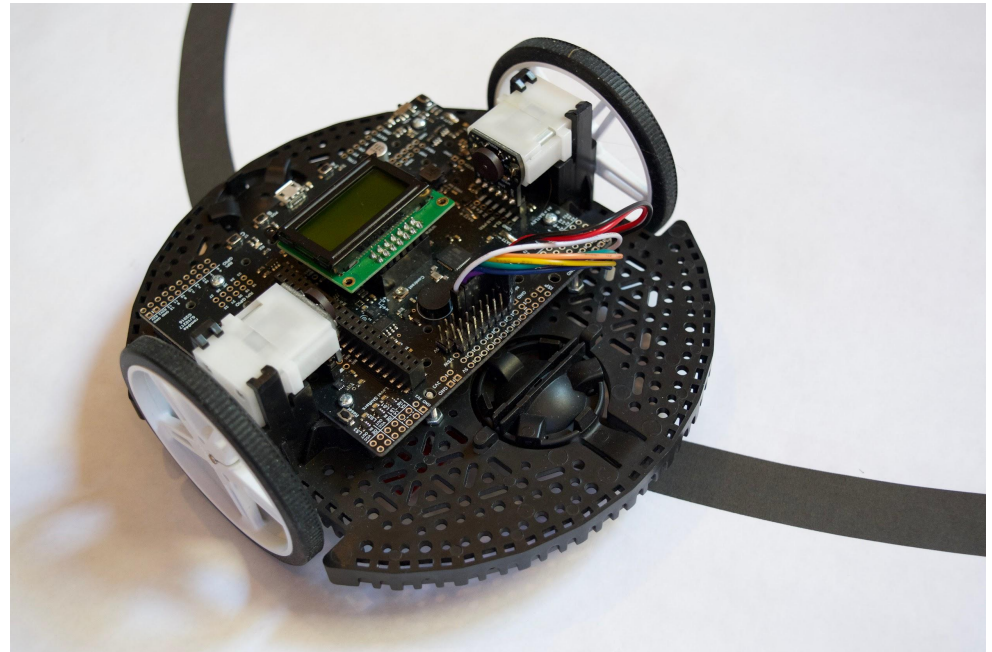
Photovore Lab

- Learn about sensors!
 - Make your robot sense light using sensors
 - Make your robot drive to the light!
- Try to beat others in both speed or distance!
 - We've had robots go from the hallway to the light bulb, detecting if they are stuck along the way!



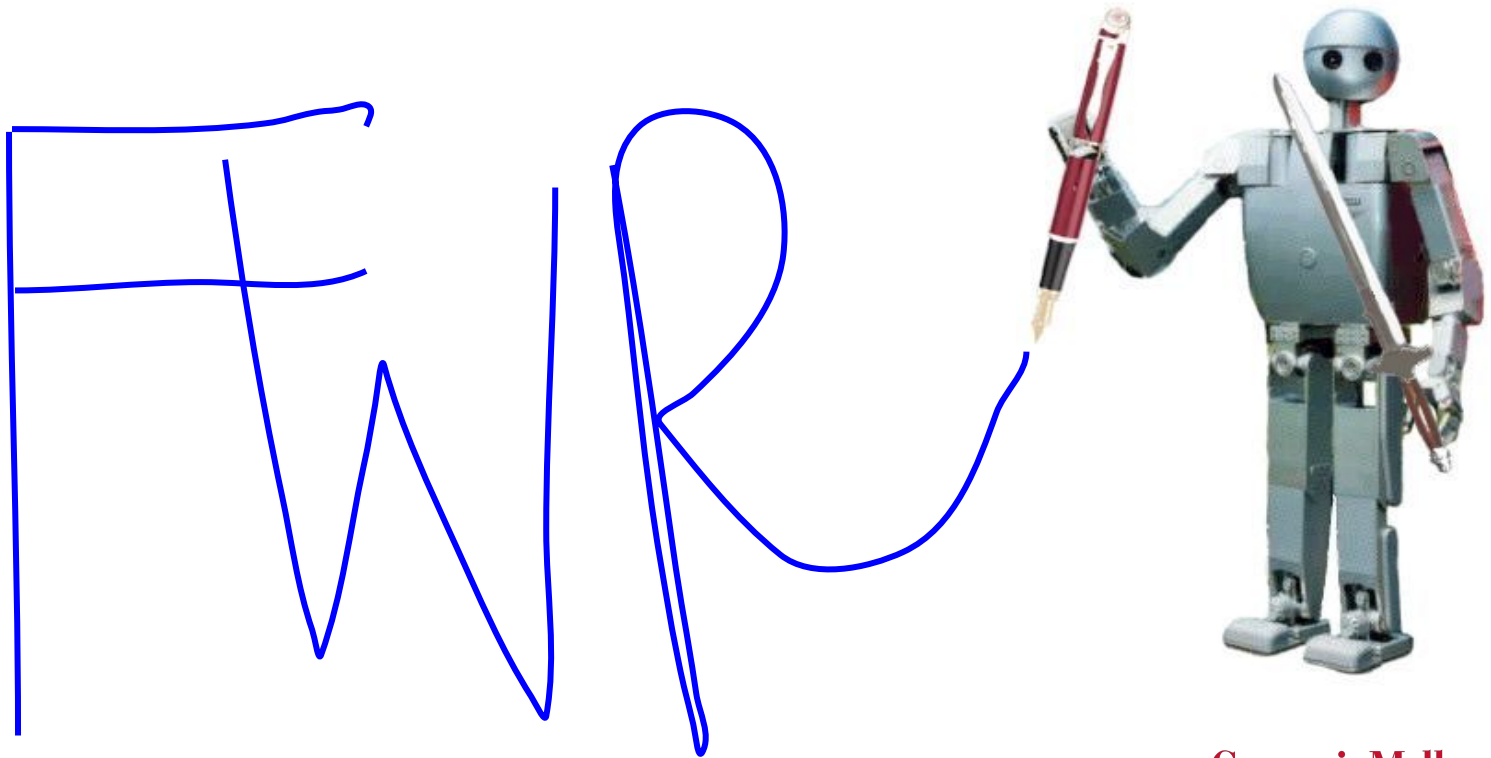
Control Lab

- Learn more complicated control algorithms!
- Make your robot find its way along a line
- Race for the fastest lap time!

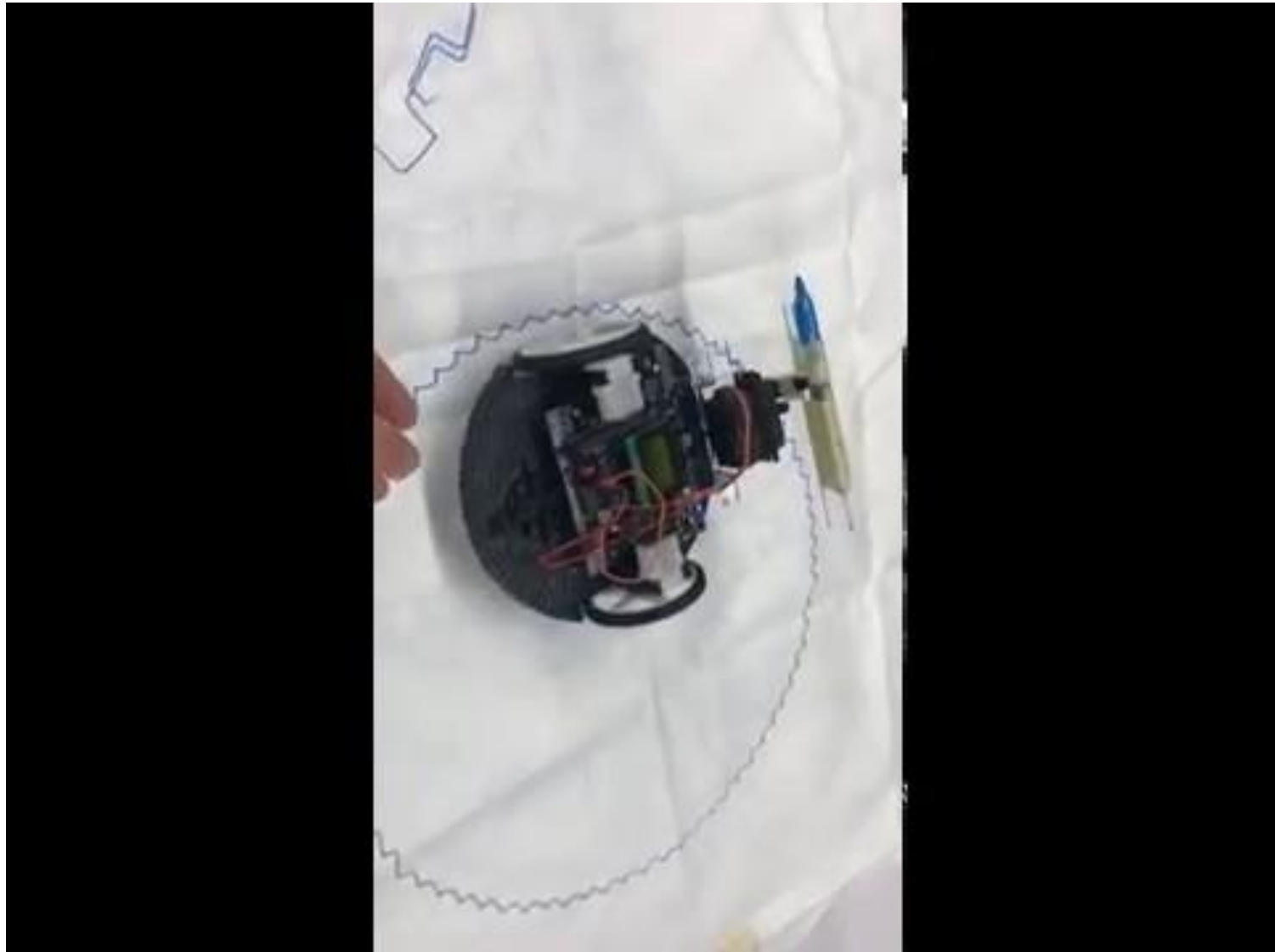


Actuation Lab

- Draw something cool with your robot and a pen!
- Or write “FWR” as fast as you can!
- Compete with other people and try to beat their time!

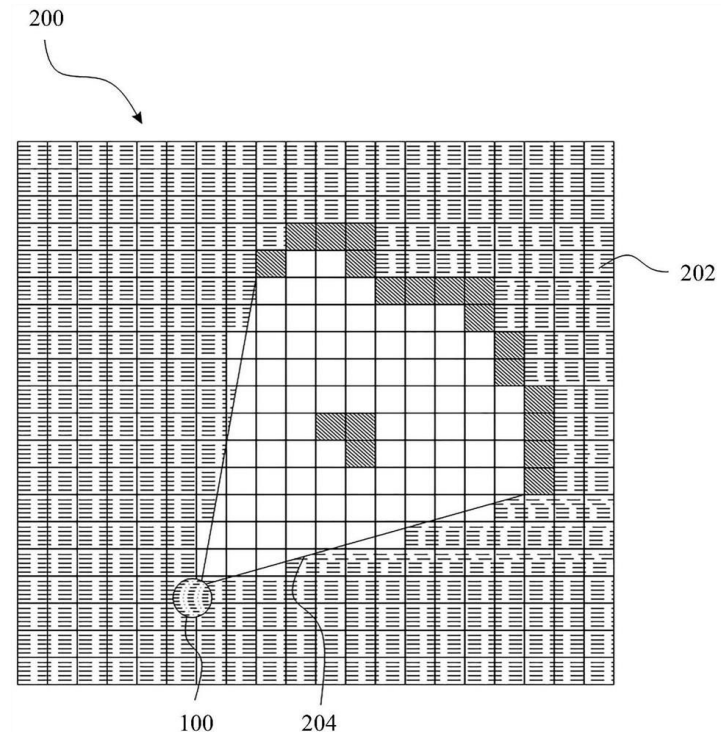


Gear Drawing Robot!



Pathfinder Lab

- Bring it all together!
 - Use sensors and code to map the area, and choose the best path.
 - A more advanced, open-ended lab that brings together programming, sensing, control, and actuation.
- Navigate with autonomy!
 - Your robot must make decisions about where to go and autonomously drive to the goal box, stopping when it arrives.
- Bonus Challenge!
 - Earn extra credit if your robot can detect when it's stuck and recover on its own!



Optional Labs

- Optional Labs Available
- If you want to learn more robotics for fun, you work on an optional lab
- You can also use these labs to “replace” a main lab if needed.



Learning Goals

Upon completion of this course, you should be able to:

- Formulate an Arduino program that controls their robot in response to inputs when given a simple task.
- Construct a circuit to connect a resistive sensor to a robot, and explain why a sensor circuit allows their robot to read a sensor.
- Utilize (and optionally, create) existing parts to create an assembly that completes a given task.
- Explain the purpose of each aspect of a PID control and write a basic PID control algorithm so the robot can complete a task effectively and efficiently.
- Describe different areas of robotics and how they are used to create functional robots.

RoboClub



About RoboClub

“Making Friends, One Robot At A Time” - 2003/2004 RoboClub T-Shirt

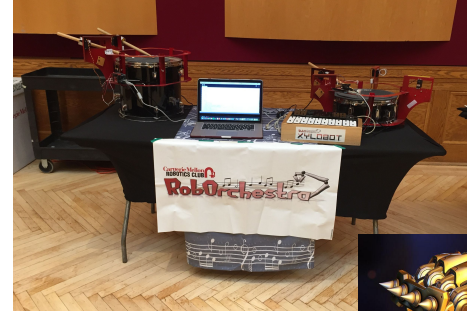
- Maker-Space/Social-Space

Why do people join?

- Learn anything or everything robotics-related
- Beginner friendly!
- Electronics Lab & Machine Shop (with training!)
- Corporate and University funding
- Social events

Club Projects

- **Robobuggy**
 - Participate in one of CMU's iconic, longest-standing traditions--but with a technological twist: Robobuggy!
- **RoboOrchestra**
 - Goal: Automate instruments to play music! Written scores & improv
- **Prosthetics**
 - Goal: Create mechanical prosthetic arms



*prosthetics did not give me a picture so i had to improvise

Club Projects Cont.

- **Combat Robotics**

- Build a BattleBots-style fighting robot together, and compete nationally!



- **3D Printer Project**

- Building a custom 3D printer!



- **Quadrupeds (NEW)**

- Work on designing, programming, and building a quadrupedal robot!



- **StuCo!**

- 98-012: This Class!
- 98-341: Build Your Own Breadboard Computer (also fun, would recommend)

*none of these projects gave me pictures so i had to improvise

Club Events

- **Red Robot Hackathon!**

- A beginner-oriented hardware hackathon for the CMU community
- Usually in the 2nd half of the Fall Semester
- Comes with food and prizes, don't miss out!
- redrobot.roboclub.org

- **Small Helpful Research Grant (SHRG)**

- Twice a year, roboclub gives out small grants for people to work on their personal projects
- This past semester, projects include a newspaper folding machine, pipette pulling machine, a 6502 based RPN calculator, and a BB8!
- Keep a lookout in the roboclub discord for more information

How to Join

Social Media:

Discord: <https://discord.gg/rnPgqXat7m>

E-mail: officers@roboclub.org

Website: roboticsclub.org

Facebook: facebook.com/groups/cmuroboclub

Instagram: [@cmuroboticsclub](https://www.instagram.com/cmuroboticsclub)

